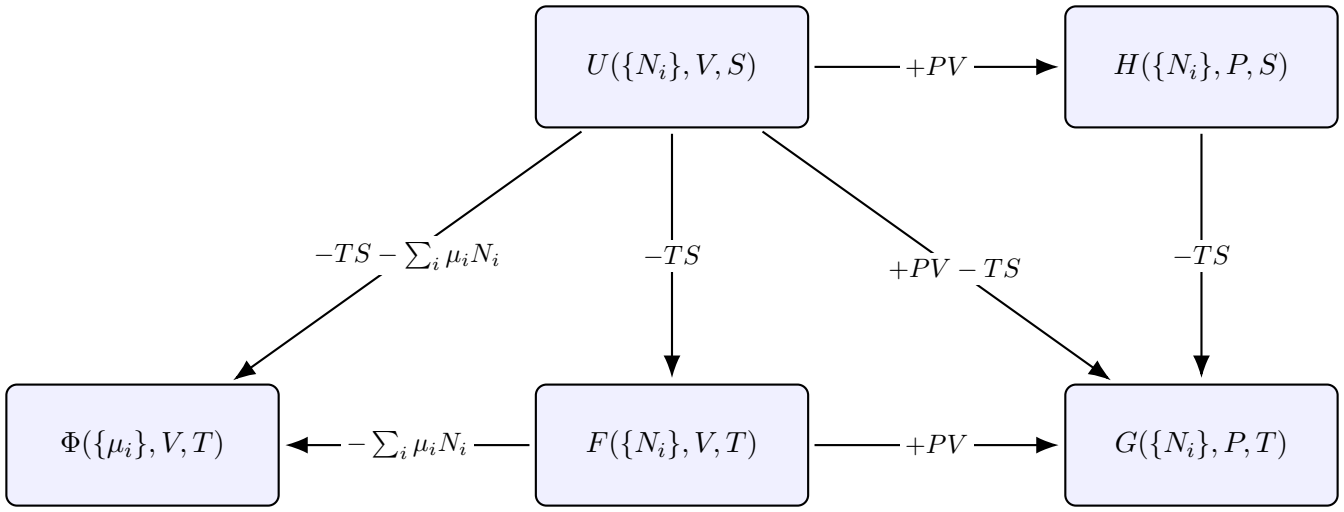
$U(\{N_i\}, V, S)$
$+PV$
$H(\{N_i\}, P, S)$
$-TS - \sum_i \mu_i N_i$
$-TS$
$+PV - TS$
$-TS$
$\Phi(\{\mu_i\}, V, T)$
$-\sum_i \mu_i N_i$
$F(\{N_i\}, V, T)$
$+PV$
$G(\{N_i\}, P, T)$